



A Comparative Analysis of the Predictive Performance of Tree-Based and Artificial Neural Network Approaches for Compressive Strength of Concrete Utilising Waste

Suhaib Rasool Wani¹ · Manju Suthar¹

Received: 31 January 2024 / Revised: 11 June 2024 / Accepted: 14 June 2024 / Published online: 9 August 2024
© The Author(s), under exclusive licence to Chinese Society of Pavement Engineering 2024

Abstract

This study attempts to develop several soft computing techniques i.e. artificial neural network (ANN), random tree (RT), random forest (RF), M5P, and reduced error pruning tree (REP tree) models for the prediction of compressive strength (CS) of concrete containing waste foundry sand. For this, a total of 343 samples were gathered from various reliable and authentic experimental research sources. Out of these, 240 (70%) samples were allocated for training purposes, simultaneously other 103 (30%) samples were designated for testing. For assessing the efficiency of each soft computing technique, statistical indicators including the Nash–Sutcliffe model efficiency coefficient (NSE), root-mean-square error (RMSE), Wilmott index (WI), mean absolute error (MAE), relative root mean square error (RRSE), coefficient of correlation (R), and relative absolute error (RAE), were calculated. The statistical indicators showed the superiority of the random forest (RF) model over the other applied models with R value of 0.94, WI value of 0.96, MAE value of 2.09, RMSE value of 2.79, RAE value of 30.89%, NSE value of 0.88, and RRSE value of 33.89% for the testing stage. The RT model has also demonstrated strong predictive capabilities compared to the ANN, REP tree, and M5P models. A sensitivity analysis revealed that the curing period was the greatest impact parameter on forecasting the compressive strength of concrete with WFS. The results of these soft computing techniques can help the construction industry by obtaining faster and more accurate analysis of material properties, particularly for WFS concrete.

Keywords Compressive strength (CS) · Random Forest (RF) · M5P · Artificial neural network (ANN) · Reduced error pruning Tree (REP Tree) · Random Tree (RT)

1 Introduction

Concrete is a prevalent substance employed in the construction sector due to its outstanding mechanical characteristics, exceptional durability features, affordability, and impressive stiffness [1]. In the current construction environment, concrete emerges as the most optimal substance for meeting construction requirements. The concrete sector is widely recognised for its substantial utilisation of natural resources, with a consumption of approximately 10

to 12 billion tons of rocks and stones [2]. Within the concrete industry, evaluating concrete quality holds immense importance. Concrete compressive strength, a fundamental mechanical property of concrete, is universally recognized as the pivotal parameter in assuring concrete quality. Mineral admixtures like silica fume (SF) and waste foundry sand (WFS) are advocated for cement manufacturing to reduce the negative environmental consequences resulting from emissions. However, the production of concrete necessitates a comprehensive assessment of the components used and accurate testing procedures to attain the desired level of compressive strength [3]. Because of technological advancement and increasing Environmental awareness, the Construction industry needs to employ high-performance materials more extensively. High-performance construction materials may exhibit superior attributes such as ductility, greater strength, and resistance to external factors. These materials not only offer

✉ Manju Suthar
manju.manni4@gmail.com; manju.e10702@cumail.in
Suhaib Rasool Wani
suhaibrwani@gmail.com

¹ Department of Civil Engineering, Chandigarh University,
NH-05, Punjab 140413, India

environmental benefits but are less expensive alternatives to conventional building materials [4]. However, there are two significant obstacles in the way of attaining environmentally friendly manufacturing in the modern days including reducing carbon emissions and the effectual recycling of waste materials. The major source of CO₂ emissions, cement production accounts for about 7% of all CO₂ released into the environment and is a vital constituent in many products [5]. Also, the amount of waste produced in concrete is increasing due to population expansion and technological improvements. This has resulted in a lack of available landfill space and higher costs for disposal, associated with dumping, recycling, and reusing industrial waste materials, making these alternatives more appealing for waste removal [6, 7]. The only way to achieve sustainable development is to increase resource efficiency. Enhancement in this aspect can be attained by decreasing the usage of both energy and materials. Over the past decade, there has been considerable interest in recycling industrial and natural wastes to develop sustainable construction materials. For instance, materials such as waste foundry sand [8, 9], recycled glass aggregate [10–13], recycled rubber aggregate [12, 13] and recycled oyster aggregate [14] are being utilized as substitutes for conventional aggregates. Due to their affordability, increased durability, and eco-friendly features, these specific concrete technologies play a crucial role in minimizing the negative impacts on both the environmental and economic aspects of the industry [15].

Waste materials are produced in large quantities by the casting processes employed in the foundry industry. Ferrous metals like cast iron and steel, and nonferrous metals such as brass, aluminium, bronze, and copper, are frequently cast in foundries. These foundries utilise moulding and casting operations to increase operating efficiency. The material is recycled multiple times until it reaches a point where additional recycling becomes impractical. Waste foundry Sand (WFS) is the name given to this final stage. Approximately fifteen per cent of the sand utilized in foundries ultimately deteriorates, contributing accumulation of numerous metric tons of waste [16].

Waste foundry sand, a by-product of casting operations, is contributing to growing economic and environmental challenges due to the expenses associated with landfill maintenance and more stringent environmental regulations. The disposal of WFS is becoming increasingly costly due to higher transportation expenses and steep landfill taxes. Although it has historically been utilized as landfill material successfully for many years, the escalating disposal costs are now posing challenges to this practice. The disposal of WFS in landfills is becoming less favourable due to the rising expenses involved. Additionally, mounting environmental concerns have led governments to implement strict

regulations to monitor industrial waste disposal and hold industries accountable for waste remediation [17, 18].

There are numerous classifications of foundry sands according to the binder systems utilised in metal casting processes. These sands are divided into two main categories: chemically bonded sand and clay-bonded sand, which is alternatively referred to as green sand. The foundry industry has created classifications that align with the various binder systems. Quartz sand typically makes up 85–95% of the material in most foundry sands, and its particle sizes range from fine to medium. These sands play a pivotal role in the foundry industry due to their composition. It is imperative to keep in mind that the specific physical and chemical attributes of foundry sand are substantially impacted by the industrial sector from which it is procured and the casting process employed [19]. Conventional fine sand is substituted by WFS in different proportions of 5%, 10%, and 15%. WFS is used to replace 15% of the sand to improve the Concrete's strength properties. With this substitution, chloride ion penetration in the concrete is reduced but ultrasonic pulse velocity (USPV) values are increased. According to this, concrete becomes denser and more impermeable [20, 21]. The CS of the concrete significantly falls when comparing the control mix and the mixes that contain 25% and 35% of WFS. However, the CS of concrete that has clean or fresh foundry sand substituted for fine aggregate at 25% and 35% of the original amount is similar to that of the control mix [22]. WFS decreases the density and increases water absorption of the concrete mixture, thereby influencing its physical properties. Unexpectedly, 20% of conventional sand can be replaced with WFS in concrete without compromising its mechanical or physical properties [23]. After a duration of 28 days, concrete formulations incorporating foundry sand substitutions of 20% and 30% exhibited a marginal reduction in CS (1.6% and 5.7%, respectively) in comparison to the control mix [16]. The degree of this sand refilling determines how much strength is lost. However, by selecting an appropriate replacement level, it is possible to obtain sufficient strength. It is possible to produce concrete using waste foundry sand by using the right mix design [24, 25]. After 28 days, the concrete showed improved compressive strength with WFS ranging from 8.25% to 17%. However, adding WFS to 15% of the mixture did not significantly increase its strength. The failure to progress may be attributed to inadequate binding caused by a reduced water-cement gel concentration in the matrix, which is the result of smaller particulates having a larger surface area [20, 21, 26].

It is crucial to accurately model the parameters of the WFS concrete, as it allows to adjust or confirm the mixing proportions [16]. Choosing the best mixing proportions can lead to cost-effective and efficient designs. Material properties can be analysed using various methods, including computer modelling, statistical techniques, and emerging tools

like regression analysis and artificial neural networks [27]. The soft computing methods are commonly used to solve issues in construction project applications.

In modern times, soft computing techniques are increasingly employed to address complex problems and reduce estimation costs and time. However, these methods require substantial computer storage space and fast processing speeds. Artificial intelligence techniques have become widely accepted and effective in the civil engineering area in the last few decades. These methods have shown distinct successful applications in various fields of civil engineering providing solutions to various complicated problems such as California Bearing Ratio, CBR [28–31], Unconfined Compressive Strength, UCS [32–34], resilient modulus [35], Marshall stability [36, 37], concrete [20, 21, 24, 38–42], bearing capacity of foundations [43], rock mechanics [44], tunneling [45], groundwater quality [46], strength of reinforced soil [47] etc. Several techniques, including genetic programming, support vector machines, random trees, M5P, and REP Trees, are frequently employed to resolve issues with construction project applications [48–50]. Comparing and contrasting the accuracy of multiple predictive models for the CS of concrete using WFS is the main objective of this study. The effects of different mix design components, such as water (W), slump (SL), coarse aggregates (CA), fine aggregates (FA), and cement (C), will be examined. These factors include the percentage of waste foundry sand (WFS) mixed with other materials (WFS/other), the ratio of water to cement (w/c), and sand (S). ANN, M5P, RT, RF, and REP Tree are a few of the models that will be employed in this study. Whichever of the ANN, M5P, REP Tree, RT, and RF models achieve the highest level of efficiency and accuracy when forecasting the CS of concrete utilizing WFS will be selected as the best model.

1.1 The Novelty of this Study

A review of the literature declares that ANN is the most performed soft computing approach to analyse the CS of cementitious elements [51]. However, the analysis necessitated the development of a model to accurately estimate the CS of concrete incorporating with WFS. In the estimation model, five modelling approaches, namely ANN, M5P, REP Tree, RT and RF were used. The present model includes ten independent variables (i.e. W, C, SL, WFS%, WFS/other, CP, W/C, S, CA and FA) and one dependent (i.e. CS) variable. The independent variable helps the model to predict more accurate results. This paper presents the comparative evaluation of efficiencies of different analysis approaches (ANN, M5P, REP Tree, RT and RF) in forecasting the CS of concrete containing WFS. Sensitivity analysis was

performed to identify the influence of input parameters on the compressive strength of concrete improved with WFS.

2 Dataset and Methodology

This research forecasted dominant soft computing models employed on collected literature data. A total of 343 observations were collected from authentic and reputable journal papers (Table 1). 70% of the dataset was utilised in the training stage, whereas 30% of the dataset was utilised in the testing stage. Table 2 presents the CS values ranging from 13.30 to 53.5 MPa, while Table 3 presents the CS values with a standard error of 0.81 MPa, a mean of 32.00 MPa, and a standard deviation of 8.28 MPa. The central objective of this research is to determine and compare the efficiency of different models in forecasting CS using WFS. For the simulation, ten variables such as SL = Slump, FA = Fine aggregate, W = Water, CA = Coarse aggregate, WFS = waste foundry sand, S = Sand, C = Cement, WFS/other = WFS mixed with other materials, CP = Curing period, and W/C = water-cement ratio were nominated as independent variables. On the other hand, compressive strength was nominated as the dependent variable. To forecast the CS of concrete using WFS by different models including ANN, M5P, REP Tree, RT and RF.

Table 1 Dataset details

Authors	Complete data set	Compressive strength
Aggarwal and Siddique [15]	28	✓
Manoharan et al. [52]	24	✓
Reshma et al. [53]	20	✓
Prabhu et al. [16]	24	✓
Naik et al., [22]	15	✓
Singh and Siddique, [20, 21]	15	✓
Zai and Salhotra [54]	15	✓
Siddique et al. [55]	20	✓
Parashar et al. [56]	15	✓
Mavroulidou and Lawrence [57]	24	✓
Prasad et al. [58]	20	✓
Bilal et al. [59]	20	✓
Mushtaq et al. [60]	18	✓
Sowmya and Kumar [61]	10	✓
Basar and Aksoy [23]	20	✓
Thiruvengadam et al. [62]	24	✓
Kavitha et al. [63]	11	✓
Siddique et al. [64]	20	✓
Total	343	343

Table 2 Overview of various concrete mixes analysed for the modelling of compressive strength

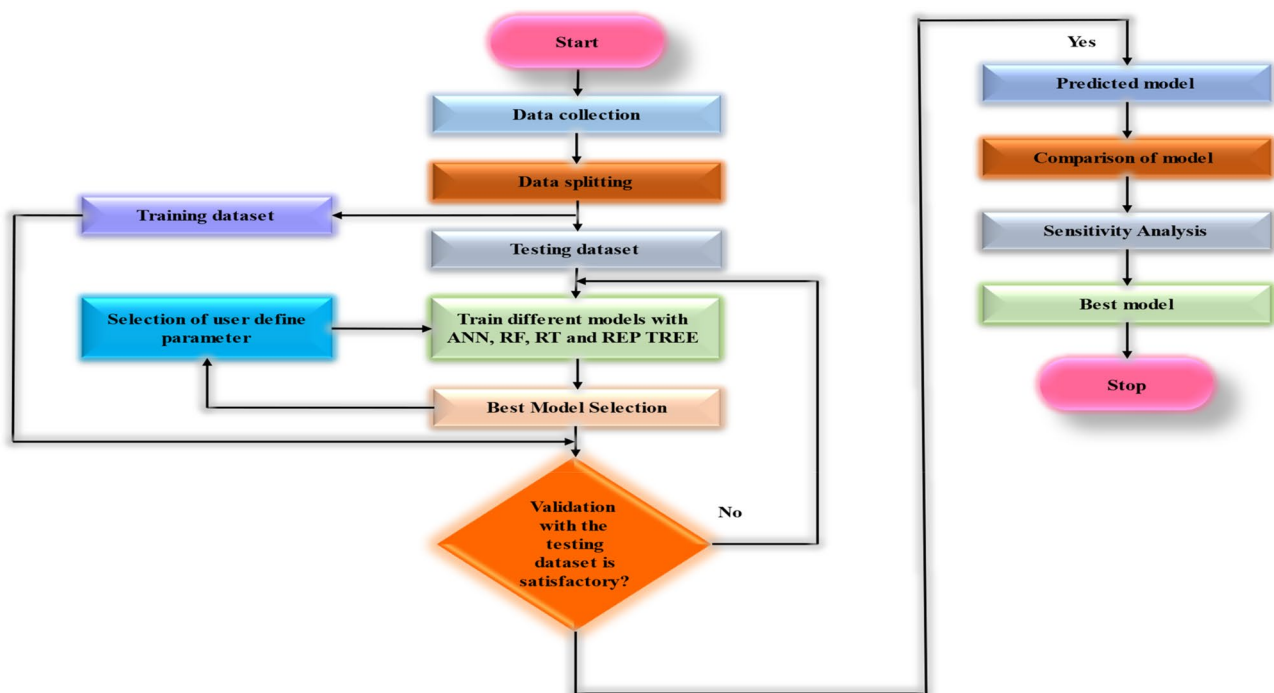
SL	CA	W	WFS (%)	S	WFS/other	C	W/C	CP (days)	FA	CS (MPa)
80–90	410	186	0–30	–377–539	0–162	372	0.5	7–365	0	19.4–39.9
30	1260	175–238.63	0–30	242–605	0–181.5	350	0.5–0.68	7–365	605	13.45–44.42
80–90	1139	189	0–20	443–554	0–111	450	0.42	28–365	0	40–53.5
29–159	1074	173	0–35	558–859	0–30	362	0.48	28–Mar	0	19.9–43.8
18–20	494	159.44–187.4	0–40	176.8–442	0–176.8	350	0.455–0.535	Jul–90	0	31–45.5
67–119	1152.25	177	0–50	308.26–616.52	0–308.26	402.37	0.44	7–180	0	16.38–37.29
70–80	1210	192	0–25	440–591	0–151	437	0.45	28–Jul	0	23.3–35.4
22–32	1309.1	187.6	0–40	0	0–261.8	436.4	0.43	Jul–91	392.7–654.5	21.5–32.4
58.8–108	590	198	0–40	910	0–364	410	0.48	Jul–90	546–910	13.3–44.2
85–115	1162.33	143.5	0–40	442.35–737.25	0–294.9	275.8	0.52	Jul–90	0	23.53–47.25
35–120	455	202	0–40	60–100	0–268.8	388	0.52	28–Jul	403.2–672	22.66–34.07
25–90	946.4	180	0–50	380.93–761.87	0–380.93	450	0.42	Jul–90	0	20–39.7
120–210	1190–1210	181.5–218	0–100	0–605	0–605	397–403.5	0.45	28–Jul	0	25.5–50.5
80–100	1235	197	0–25	453–604	0–151	394	0.5	28–Jul	0	16.4–26.6
80–90	1165	195	0–20	456–569	0–113	390	0.5	7–365	0	20–45.2
75–100	482.2	157.78	0–50	377–683.4	0–341.7	394.32	0.4	28–Jul	241.6	31–47
0	1082.72	146–208	0–30	455–649.72	64.9–649.72	291.6–416.64	0.5	28–Jul	83.32–416.64	17–37
69–117	1325	148	0–50	0	0–373	370	0.4	28	298.4–746	38–49.7

SL Slump; FA Fine aggregate; W Water; CA Coarse aggregate; WFS waste foundry sand; S Sand; C Cement; CP Curing period; W/C water cement ratio; CS Compressive strength.

Table 3 Statistical characteristics of a dataset

	SL	CA	W	WFS (%)	S	WFS/other	C	W/C	CP	FA	CS (MPa)
Training Dataset ($n = 240$)											Output
Standard Error	2.86	19.85	1.24	1.16	14.19	8.01	2.92	0	5.33	16.61	0.56
Median	80	1152.25	186	20	485	109.5	394	0.48	28	0	32.15
Maximum	210	1325	238.63	100	910	649.72	450	0.68	365	910	53.5
Minimum	0	410	143.5	0	0	0	275.8	0.4	3	0	13.3
Kurtosis	1.01	-0.58	0.31	4.55	0.26	2.77	0.18	1.27	8.4	-0.04	-0.62
Mean	74.52	1012.53	182.61	20.33	444.92	135.23	385.97	0.47	53.48	163.28	32.15
Standard Deviation	44.37	307.66	19.27	18.11	219.93	124.23	45.26	0.05	82.6	257.41	8.74
Testing Dataset ($n = 103$)											Output
Standard Error	4.25	29.88	1.84	1.64	21.17	12.186	4.43	0	9.33	25.42	0.81
Median	80	1162.33	186	20	484	130	394	0.5	28	0	31.6
Maximum	210	1325	238.63	70	910	649.72	450	0.68	365	910	49.8
Minimum	0	410	143.5	0	0	0	275.8	0.4	3	0	16.38
Kurtosis	1.11	-0.34	0.49	0.48	0.52	1.71	0.45	1.53	6.22	-0.02	-0.75
Mean	76.83	1022.4	182.44	20.77	461.43	142.86	384.55	0.47	57.23	160.39	32
Standard Deviation	43.15	303.31	18.71	16.69	214.86	123.68	44.98	0.05	94.76	258.03	8.28

SL Slump; FA Fine aggregate; W Water; CA Coarse aggregate; WFS waste foundry sand; S Sand; C Cement; CP Curing period; W/C water cement ratio; CS Compressive strength.

**Fig. 1** Methodology flow chart

3 Model Architecture

The present study employed five soft computing techniques, such as ANN, M5P, REP Tree, RT, and RF. The flowchart of the following approaches is presented in

Fig. 1 whereas Fig. 2 presents the correlation matrix for the entire dataset. There was not a significant improvement in the strength after 15% waste foundry sand, which might be related to an increase in fine particle surface area that would have reduced the amount of water cement in the

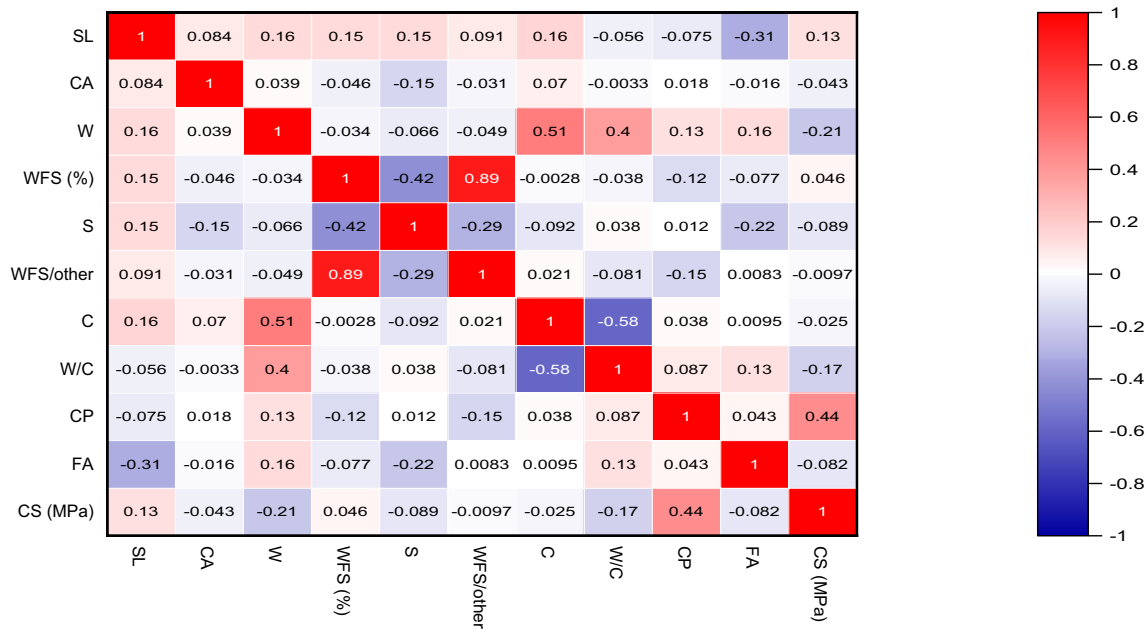
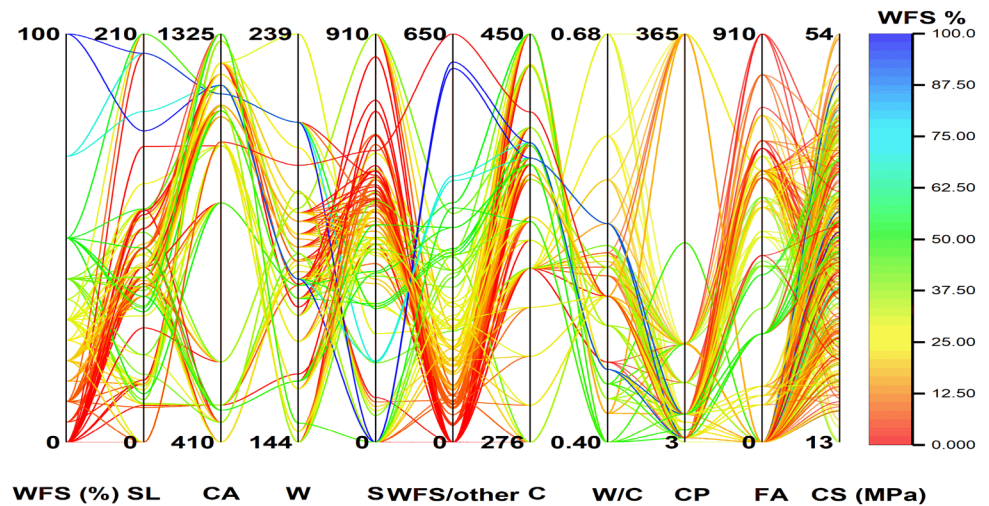


Fig. 2 Correlation matrix of the entire collected dataset

Fig. 3 The parallel correlation among the variable, WFS% and CS



matrix and resulted in insufficient binding [20, 21, 26] which is illustrated in Fig. 3 by the parallel correlation among the variable, WFS% and CS. The following section provides a brief summary of the models utilized in this study:

3.1 Artificial Neural Network (ANN)

Since the brain operates differently from traditional digital computers, artificial neural networks have been constructed. With ANNs, decentralised, highly parallel computers are combined with concealed input and output layers. Since the

intended output and input parameters form the network's total number of neurons, the number of input and output neurons is defined by this total number [65]. The network's input and output data are utilised to train the ANN model in (MATLAB). To minimise errors, the model's weights and biases are adjusted iteratively during this process. To make this change, compare the desired and expected outcomes [66]. The methodology used in this study involved randomly partitioning the dataset into 70% for training and 30% for testing purposes. Finding the best hyperparameters for the ANN model required making methodical changes to the number of hidden neurons. Utilizing the

Levenberg–Marquardt function a well-respected backpropagation method recognized for its efficiency weights and biases was optimized. It is generally advised to use this technique for supervised training methods [67].

The three primary layers of the artificial neural network are depicted in Fig. 4 Which is given below:

3.2 Random Forest (RF)

Breiman developed the incredibly adaptable random forest technique, which is commonly used to address a diversity of challenging engineering obstacles [68]. One flexible method in soft computing is the Random Forest algorithm. There are several decision trees involved, all of which were built with different training data subsets. Each tree utilizes data subsets, and the nodes are split based on the best candidates from randomly chosen variables. This approach reduces overfitting, enhancing the model's ability to generalize. The regression-supporting R programming language (version 4.0.2) was used to develop the RF model in this

situation. Breiman's initial random forest method served as the foundation for the model [69]. This approach incorporates extra features such as quantile regression forests and very random trees. Moreover, it surpasses the speed of the unique algorithm. The random forest generator, version 0.12.1, of the “Ranger” package, was used to build the model [70]. Figure 5 illustrates the network diagram of the Random Forest-based model. The expected response variations were used by the program to separate the trees into the forests. Within each forest, a total of 500 trees of the same type were produced using the projected response variances as the tree type. Numerous factors can contribute to RF overfitting. As previously said, if the training data is strongly fitted (resulting in deep trees), generalization is hampered. To reduce overfitting, the algorithm's hyper-parameters were altered. The split formula that establishes the divisions between trees, the total number of trees, the depth to which a tree can reach, and the characteristics that are considered for each tree are all encompassed by these parameters.

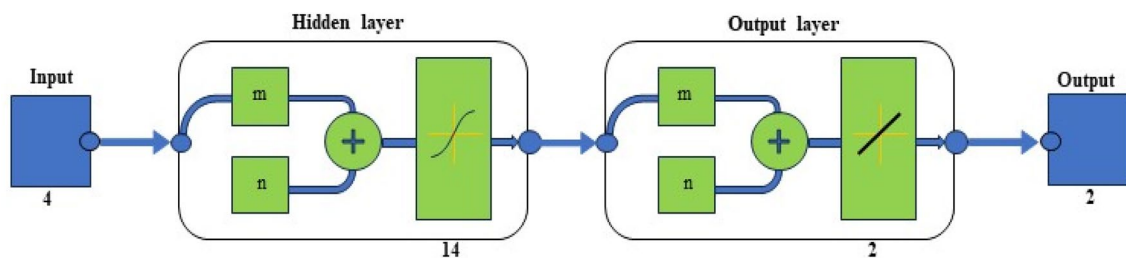
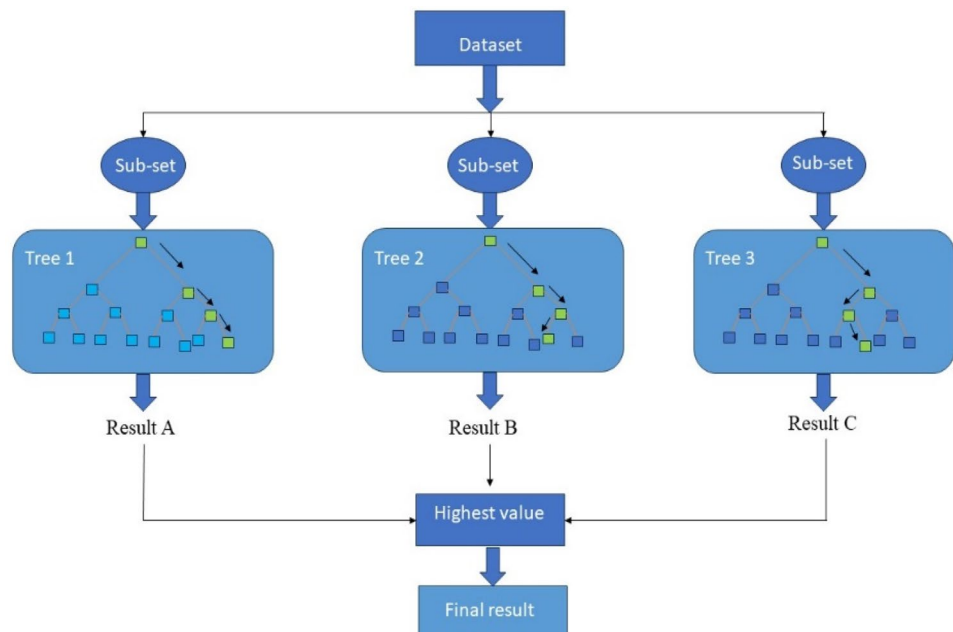


Fig. 4 Structure of ANN

Fig. 5 Random Forest flow network diagram



3.3 Reduced Error Pruning Tree (REP Tree)

This approach to decision tree learning constructs decision trees by evaluating data gain or variance. Through methods like reduced error pruning and a back-fitting algorithm, these trees are further refined [71]. The tree is pruned based on the mean square error resulting from its predictions, serving as the criterion for refinement. Simply "REP Tree" is a method for understanding decision trees rapidly. It constitutes a decision tree through the assessment of information enhancements and the reduction of variation. Information gain is utilized as the splitting criterion in the REP Tree, a rapid decision tree learning system. Reduced error pruning techniques are then applied to augment the tree structure. Only one sorting operation is performed on numerical attribute values, and C4.5's fractional instances technique is utilized to manage missing values. [72]. A visual representation of the model based on the Reduced Error Pruning (REP) technique is depicted in Fig. 6.

3.4 M5P Model

The model trees technique incorporates a segmented linear approximation that is specific to each form of data so as to generate a representation of the data. They keep the conventional decision tree form, but rather than using discrete class labels at the terminal nodes, they use linear functions [73]. In numerical prediction, a model tree is employed where each terminal node contains a linear regression model. Every leaf node occupies a model that forecasts the class value of occurrences that will eventually influence that node [74]. Model Trees offer increased prediction accuracy and compactness over traditional regression trees [75].

3.4.1 M5P Tree Algorithm

Model trees, despite their apparent simplicity, are exceptionally efficient and accurate methods for detecting patterns and correlations within huge data sets [76]. Quinlan introduced an innovative decision tree method called M5 prime, specifically designed to tackle intricate and nonlinear challenges in massive datasets, while taking

crucial input factors into account [77]. Tree construction, tree trimming, and tree smoothing are the three basic processes of building an M5 tree, as shown in Fig. 7. Standard Deviation Reduction (SDR) is a metric that is mostly used while building a tree.

The Eq. 1 of SDR is defined as:

$$\text{SDR} = S_d(M) - \sum_i \frac{M_i}{|M|} \times S_d(M_i) \quad (1)$$

In this context, "M" indicates a set of instances, while "M_i" indicates the particular subgroup of instances that resulted from partitioning the initial set using a certain collection of variables (e.g., attributes). $S_d(M)$ denotes the set's Standard Deviation, while $S_d(M_i)$ denotes the subset's standard deviation, serving as a measure of the error inside that subset.

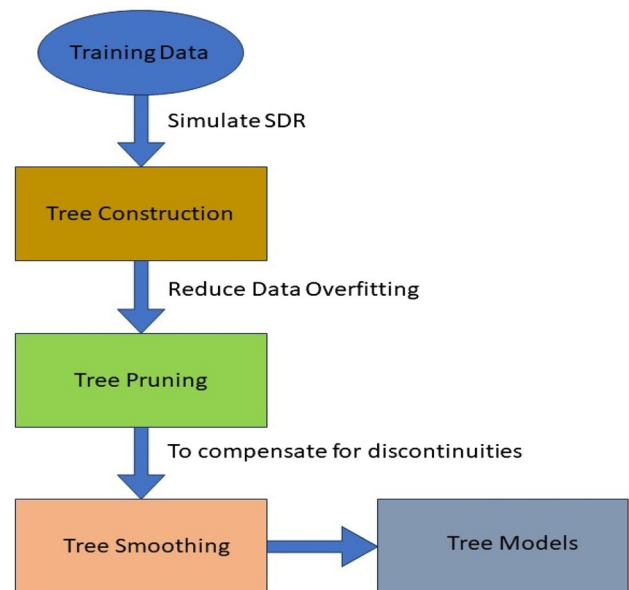
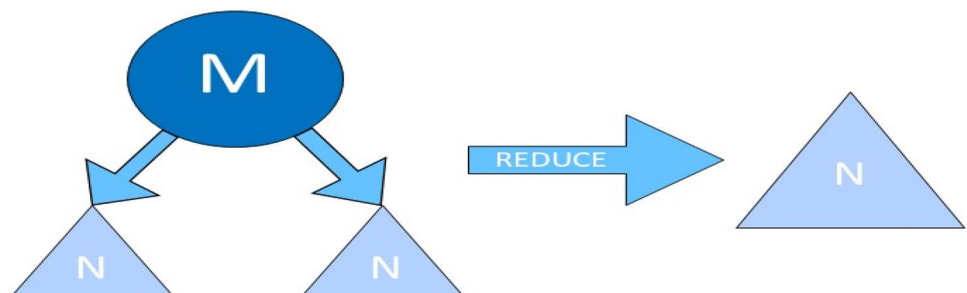


Fig. 7 M5P model tree flowchart

Fig. 6 Demonstrate a REP-based model



3.5 Random Tree (RT)

The random tree model, introduced by David Aldous, is a flexible decision/regression model that can effectively manage both classification and regression tasks, eliminating the necessity for pruning techniques [78]. Like Bagging and Random Forest, Random Tree operates as a supervised soft computing classifier functioning as an ensemble learning algorithm [69]. Every decision tree leaf in random trees contains a localized linear model that has been optimized for the particular subspace it represents. Two randomization strategies are used to increase diversity and boost each decision tree's performance. The training data is sampled initially with a replacement for every tree, which bears a resemblance to the methodology utilised in bagging. In the progression of the tree, the optimal split is ascertained by choosing a random subset of all node-level attributes. This method uses the random split selection strategy to generate a uniform ridge value setting across all leaves in order to improve the optimisation process and encourage balanced trees [79] which is presented in Fig. 8.

4 Performance Criterion Evaluation

A variety of statistical indicators were utilised to analyse both training and testing datasets to determine the accuracy of the model that has been developed. The statistical

performance evaluation metrics used in this analysis are RRSE, RAE, NSE, WI, MAE, RMSE, and R.

4.1 Coefficient of Correlation (R)

The statistical reliability of the relationship between two variables is assessed using the R-value. This coefficient, designated as R [80], is computed as follows using the provided Eq. 2.

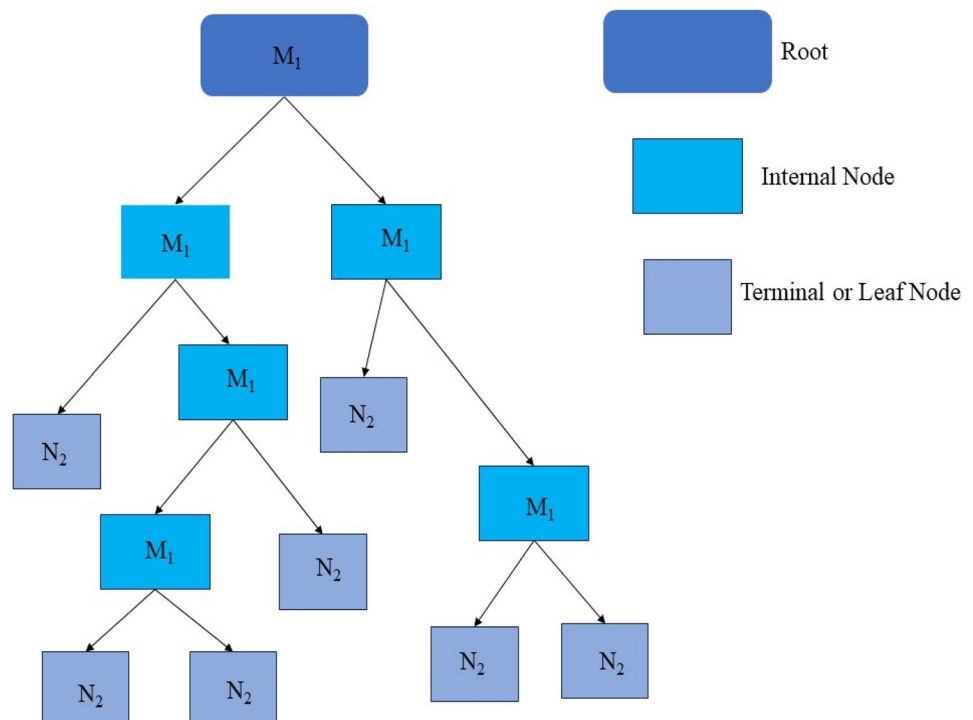
$$R = \frac{\sum_{i=1}^t CS_O CS_P - (\sum_{i=1}^t CS_O)(\sum_{i=1}^t CS_P)}{\sqrt{t(\sum_{i=1}^t CS_O^2) - (\sum_{i=1}^t CS_O)^2} \sqrt{t(\sum_{i=1}^t CS_P^2) - (\sum_{i=1}^t CS_P)^2}} \quad (2)$$

In this context, CS_O stands for compressive strength observation, CS_P stands for compressive strength prediction, and 't' is for the observational count. The range of correlation coefficients (R) is from -1 to 1. Between variables CS_O and CS_P (observed and forecasted values), the value of 1 indicates a perfect positive correlation, whereas a value of -1 indicates a perfect negative correlation. No linear association between variables CS_O and CS_P is indicated by a value of 0.

4.2 Wilmott Index (WI)

Wilmott created the agreement index in 1981 to quantify the level of inaccuracy of the model. It ranges from 0 to 1. This measure illustrates the relationship between the MSE and the potential error. A number of 0 directs no agreement, while

Fig. 8 Basic structure of decision tree



a rating of 1 indicates a perfect match. The agreement index can detect variations between simulated and observed means as well as additive and proportional variances. Nevertheless, because it uses squared differences in its derivation, the agreement index (WI) is very susceptible to extreme values.

$$WI = 1 - \frac{\sum_{i=1}^t (CS_O - CS_P)^2}{\sum_{i=1}^t (|CS_P - \overline{CS_O}| + |CS_O - \overline{CS_O}|)^2} \quad (3)$$

In this context, CS_O stands for compressive strength observation, and CS_P stands for compressive strength prediction. Additionally, $\overline{CS_O}$ stands for the mean of observations and 't' is for the observational count.

4.3 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the difference between actual and forecasted values. Utilising the provided equation no. 4 by which the determination is achieved (Kharb et al., 2021):

$$MAE = \frac{1}{t} \left(\sum_{i=1}^t |CS_P - CS_O| \right) \quad (4)$$

In this context, CS_O stands for compressive strength observation, and CS_P stands for compressive strength prediction. Additionally, 't' is for the observational count.

The MAE value has a range of 0 to ∞ . Perfect or nearly perfect agreement between the actual and expected values is indicated by a number of 0.

4.4 Nash–Sutcliffe Efficiency (NSE)

The Nash–Sutcliffe efficiency (NSE) indicator shows the predictive power of the applied model and indicates its effectiveness. This equation is used to calculate NSE:

$$NSE = 1 - \frac{\sum_{i=1}^t (CS_O - CS_P)^2}{\sum_{i=1}^t (CS_O - \overline{CS_O})^2} \quad (5)$$

In this context, CS_O stands for compressive strength observation, CS_P stands for compressive strength prediction, and 't' is for the observational count. Additionally, $\overline{CS_O}$ stands for the mean of observations.

The range of the NSE values is between $-\infty$ and 1. The accuracy of the model is in line with the observed data mean when the NSE is 0. An NSE score of 1, on the other hand, indicates complete agreement that is, no error between the data that the model forecasts and the observed data.

4.5 Root-Mean-Square-Error (RMSE)

The square root of the mean squared error (MSE) constitutes the definition of the RMSE. The square root of the variance between observed and forecasted values is computed. This equation is used to calculate the RMSE value:

$$RMSE = \sqrt{\frac{1}{t} \left(\sum_{i=1}^t (CS_P - CS_O)^2 \right)} \quad (6)$$

In this case, CS_O stands for compressive strength observation, CS_P for compressive strength prediction, and 't' is for the observational count. The RMSE value falls between 0 and ∞ . A value of 0 signifies an ideal and perfect model, accurately forecasting the observed values.

4.6 Relative Root Mean Square Error (RRSE)

The statistical metric is utilized to estimate the accuracy of the model by the relative root mean square error. It measures the relative difference between the estimated and observed values as a modification of the RMSE. It is represented by:

$$RRSE = \sqrt{\frac{\sum_{i=1}^t (CS_O - CS_P)^2}{\sum_{i=1}^t |CS_O - \overline{CS_P}|^2}} \quad (7)$$

While CS_P represents compressive strength expectation in this present study, $\overline{CS_O}$ represents compressive strength perception. Also, $\overline{CS_O}$ addresses the mean of the perceptions, and "t" is for the observational count.

4.7 Relative Absolute Error (RAE)

The model divides the sum of absolute error by the sum of absolute deviation from the observed value with their mean to determine its relative absolute error for prediction. So is addressed by the numerical condition:

$$RAE = \frac{\sum_{i=1}^t |CS_O - CS_P|}{\sum_{i=1}^t |CS_O - \overline{CS_O}|} \quad (8)$$

While CS_P represents compressive strength expectation in this present study, CS_P represents compressive strength perception. Also $\overline{CS_O}$ addresses the mean of the perceptions and 't' is for the observational count.

5 Result and Discussion

The outcomes of the various models engaged in this study to forecast the value of compressive strength utilised with waste foundry sand have been exhibited in this section:

5.1 Assessment of M5P-Based Model

The statistical parameters utilised to assess the efficiency of the M5P-based model are illustrated in Table 4 which indicates that the M5P-based model for forecasting compressive strength is satisfactory but not remarkable with R as 0.86, 0.82, RMSE as 4.38, 4.62, MAE as 3.47, 3.75, RAE as 48.87%, 55.53%, RRSE as 50.29%, 56.13%, WI as 0.74, 0.68 and NSE as 0.91, 0.89 for both the training as well as testing stages respectively.

The efficiency of the M5P-based model in forecasting the CS of concrete containing WFS is illustrated in Fig. 9 during both the training as well as testing stages. The agreement line and the relative error variation are outlined for both the training and testing stages.

The results also exhibit that the forecasted values of the compressive strength during the training stage lie between $\pm 36.6\%$ and $\pm 33.3\%$ for the testing stage, which is significantly high in range. The user-defined parameters are also shown in Table 5 which is determined through the hit and trial method. As anticipated, in comparison with the training stages, the accuracy of the M5P-based model exhibited a decline during the testing stage.

5.2 Assessment of REP Tree-Based Model

The statistical parameters utilised to assess the efficiency of the REP tree-based model are illustrated in Table 4 which

indicates that the REP tree-based model for forecasting compressive strength is satisfactory and better than the M5P-based model but not remarkable with R as 0.89, 0.86, RMSE as 3.84, 4.22, MAE as 2.70, 3.09, RAE as 38.02%, 45.66%, RRSE as 44.02%, 51.21%, WI as 0.92, 0.94 and NSE as 0.80, 0.73 for both the training as well as testing stages respectively.

The efficiency of the REP tree-based model in forecasting the CS of concrete containing WFS is illustrated in Fig. 10 during both the training as well as testing stages. The agreement line and the relative error variation are outlined for both the training and testing stages.

The results also exhibit that the forecasted values of the compressive strength during the training as well as testing stage lie between $\pm 35\%$, which is a significantly high range but better than the M5P-based model. The user-defined parameters are also shown in Table 5 which is determined through the hit and trial method. As anticipated, in comparison with the training stage, the accuracy of the M5P-based model exhibited a decline during the testing stage.

5.3 Assessment of ANN-Based Model

The statistical parameters utilised to assess the efficiency of the ANN-based model are illustrated in Table 4 which indicates that the ANN-based model for forecasting compressive strength is good and better than the M5P-based model and REP tree-based model with R as 0.93, 0.90, RMSE as 3.35, 3.64, MAE as 2.60, 2.86, RAE as 36.57%, 42.37%, RRSE as 38.44%, 44.23%, WI as 0.96, 0.94 and NSE as 0.85, 0.80 for both the training as well as testing stages respectively.

The efficiency of the ANN-based model in forecasting the CS of concrete containing WFS is illustrated in Fig. 11

Table 4 Metrics for soft computing model evaluation throughout the training and testing stages

	R	MAE	RMSE	RAE	RRSE	WI	NSE
	Training ($n = 240$)						
M5P	0.86	3.47	4.38	48.87%	50.29%	0.74	0.91
REP tree	0.89	2.7	3.84	38.02%	44.02%	0.94	0.8
ANN	0.93	2.6	3.35	36.57%	38.44%	0.96	0.85
RT	0.99	0.04	0.1	0.69%	1.16%	1	1
RF	0.99	0.91	1.24	12.83%	14.24%	0.99	0.85
	Testing ($n = 103$)						
	R	MAE	RMSE	RAE	RRSE	WI	NSE
M5P	0.82	3.75	4.62	55.53%	56.13%	0.68	0.89
REP tree	0.86	3.09	4.22	45.66%	51.21%	0.92	0.73
ANN	0.9	2.86	3.64	42.37%	44.23%	0.94	0.8
RT	0.89	2.82	3.96	41.75%	48.05%	0.94	0.76
RF	0.94	2.09	2.79	30.89%	33.89%	0.96	0.88

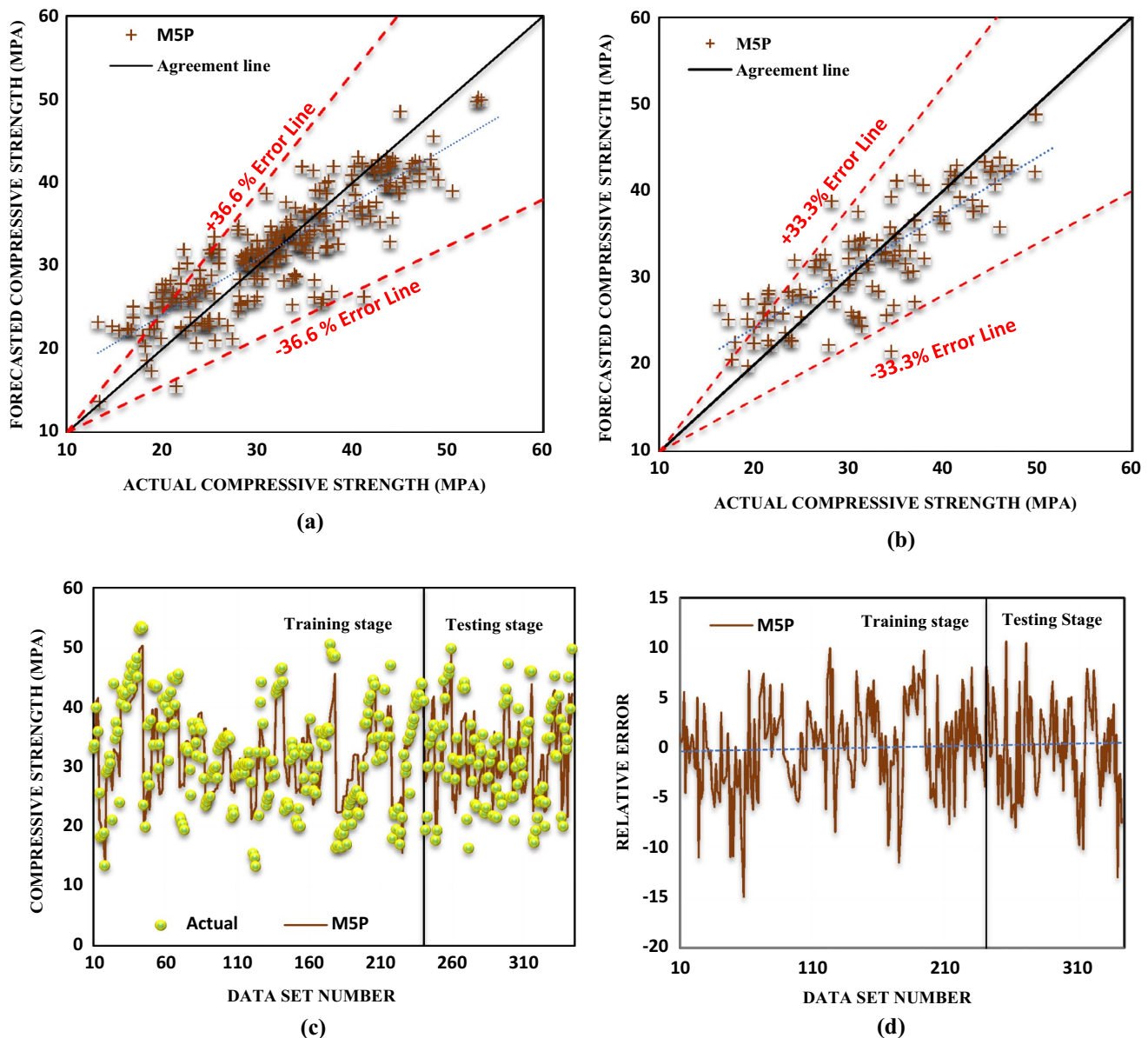


Fig. 9 Agreement plot of actual as well as forecasted values of M5P-based model: **a** direct correlation for the training stage ($n=240$) **b** direct correlation for the testing stage ($n=103$) **c** a head-to-head

agreement was executed concerning both the training and assessment stages ($n=343$) **d** training and testing stages' relative error ($n=343$)

Table 5 Optimal values for user-defined parameters

Models used	User-defined parameters (constants)
M5P	$M=4$
REP tree	$M=2, V=0.001, N=3, S=2, L=-1, I=0.0$
ANN	$HL=9, LR=0.3, M=0.2, TT=1000$
RT	$K=0, M=1, V=0.001, S=1, N=0$
RF	$K=4, I=100, \text{Seed}=1$

during both the training as well as testing stages. The agreement line and the relative error variation are outlined for both the training and testing stages.

The results also exhibit that the forecasted values of the compressive strength during the training as well as testing stage lies between $\pm 25\%$, which is a significantly good in range and better than the M5P and REP tree-based model. The user-defined parameters are also shown in Table 5 which is determined through the hit and trial method. As anticipated, in comparison with the training stage, the accuracy of

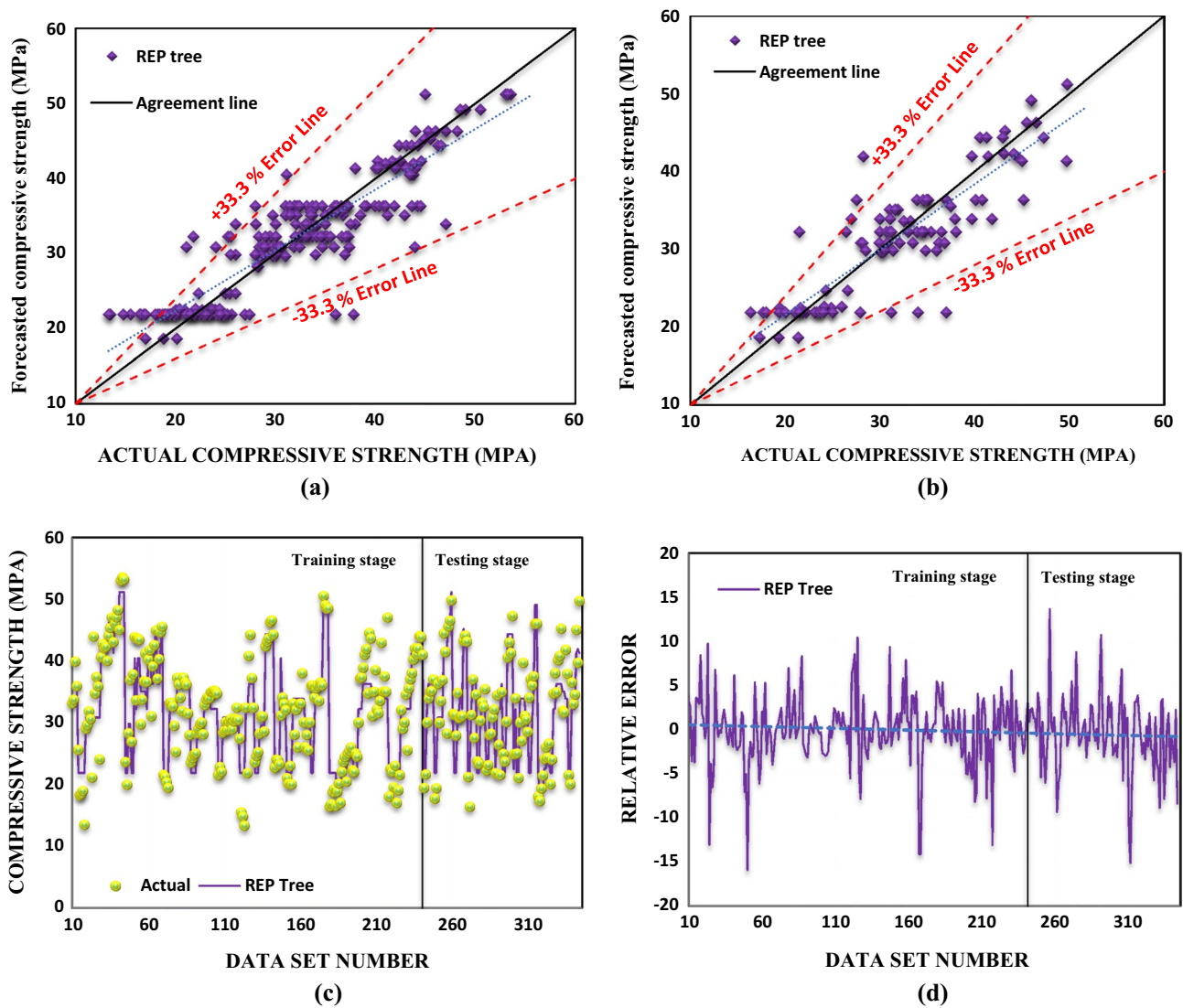


Fig. 10 Agreement plot of actual as well as forecasted values of REP tree-based model: **a** direct correlation for the training stage ($n=240$) **b** direct correlation for the testing stage ($n=103$) **c** a head-to-head

agreement was executed concerning both the training and assessment stages ($n=343$) **d** Training and testing stages' relative error ($n=343$)

the ANN-based model exhibited a decline during the testing stage.

5.4 Assessment of RT-Based Model

The statistical parameters utilised to assess the efficiency of the RT-based model are illustrated in Table 4 which indicates that the RT-based model for forecasting CS is finest and better than M5P, REP tree and ANN-based models with R as 0.99, 0.89, RMSE as 0.10, 3.96, MAE as 0.04, 2.82, RAE as 0.69%, 41.75%, RRSE as 1.16%, 48.05%, WI as 1,

0.94 and NSE as 1, 0.76 for both the training as well as testing stages respectively.

The efficiency of the RT-based model in forecasting the CS of concrete containing WFS is illustrated in Fig. 12 during both the training as well as testing stages. The agreement line and the relative error variation are outlined for both the training and testing stages. The user-defined parameters are also shown in Table 5 which is determined through the hit and trial method.

The results also exhibit that the forecasted values of the compressive strength during the training stage lie between $\pm 11.6\%$ and $\pm 25\%$ for the testing stage, which is

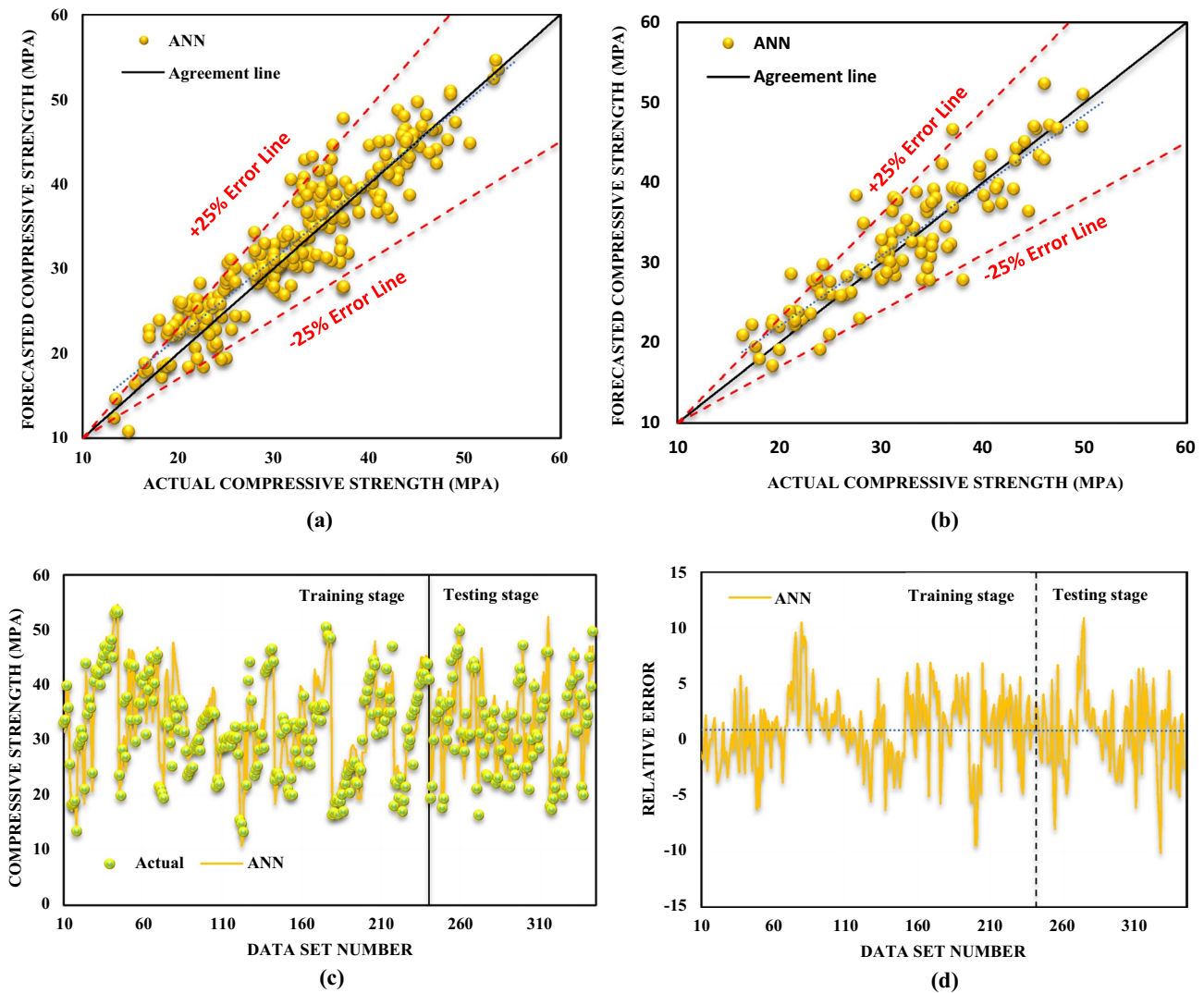


Fig. 11 Agreement plot of actual as well as forecasted values of ANN-based model: **a** direct correlation for the training stage ($n=240$) **b** direct correlation for the testing stage ($n=103$) **c** a head-

to-head agreement was executed concerning both the training and assessment stages ($n=343$) **d** Training and testing stages' relative error ($n=343$)

a significantly very good range and better than M5P, REP tree and ANN-based models. Results of training phase of the RT model show overfitting as the model fits the training data too closely (Fig. 12 a, d). To mitigate overfitting during the training phase, cross-validation techniques such as K-Fold Cross-Validation and Leave-One-Out Cross-Validation are commonly utilized [81].

5.5 Assessment of RF-Based Model

The statistical parameters utilised to assess the efficiency of the RF-based model are illustrated in Table 4 which

indicates that the RF-based model for forecasting compressive strength is the best and outperforms other models that have been implemented in terms of forecasting compressive strength with R as 0.99, 0.94, RMSE as 1.24, 2.79, MAE as 0.91, 2.09, RAE as 12.83%, 30.89%, RRSE as 14.24%, 33.89%, WI as 0.99, 0.96 and NSE 0.85, 0.88 for both the training as well as testing stages respectively.

The efficiency of the RF-based model in forecasting the CS of concrete containing WFS is illustrated in Fig. 13 during both the training as well as testing stages. The agreement line and the relative error variation are outlined for both the training and testing stages.

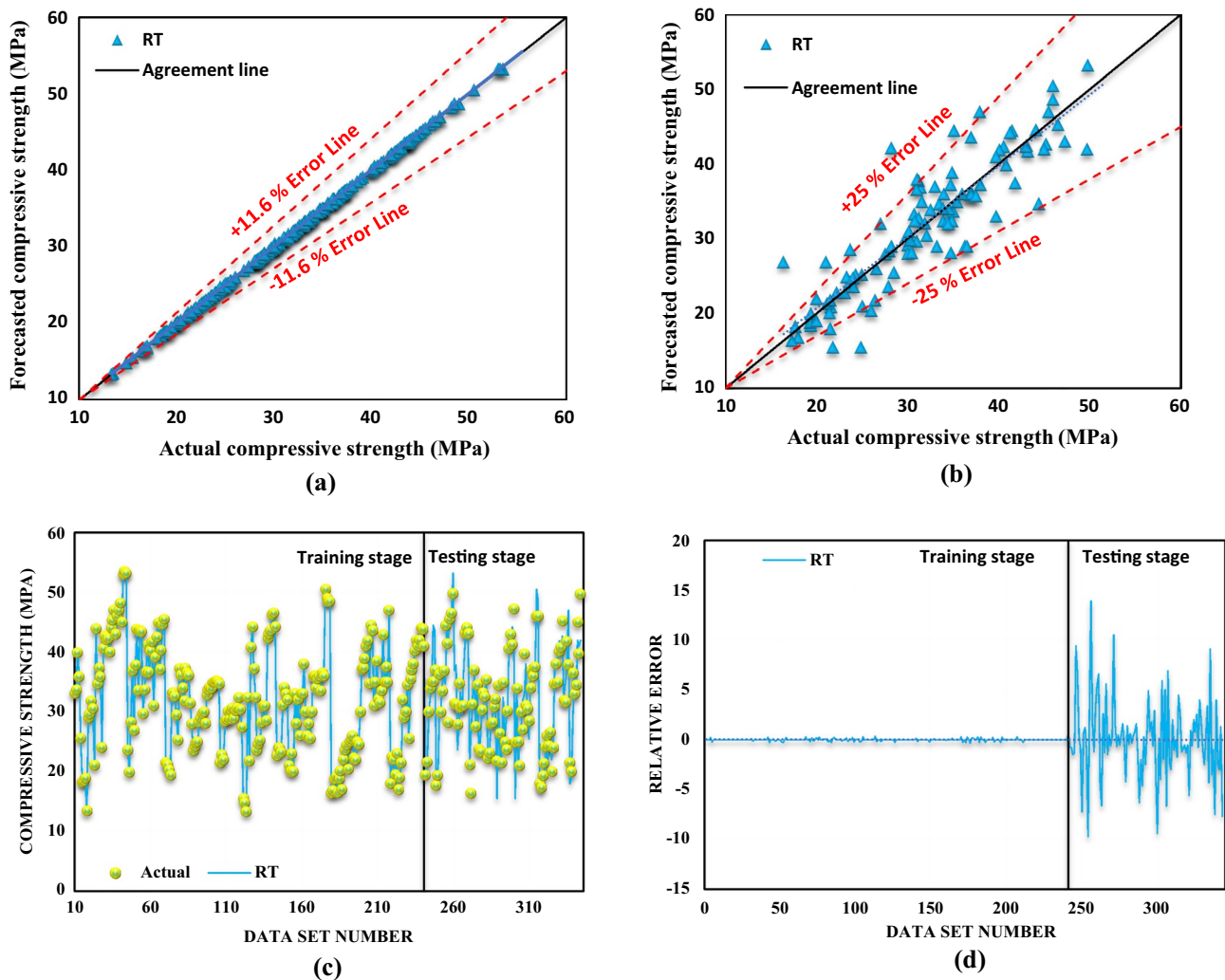


Fig. 12 Agreement plot of actual as well as forecasted values of MSP-based model: **a** direct correlation for the training stage ($n=240$) **b** direct correlation for the testing stage ($n=103$) **c** A head-to-head

agreement was executed concerning both the training and assessment stages ($n=343$) **d** training and testing stages' relative error ($n=343$)

The results also exhibit that the forecasted values of the compressive strength during the training stage lie between $\pm 11.6\%$ and $\pm 16.6\%$ for the testing stage, which is a significantly very good range and better than M5P, REP tree, ANN and RT based models. The user-defined parameters are also shown in Table 5 which is determined through the hit and trial method. As anticipated, in comparison with the training stage, the accuracy of the RF-based model exhibited a decline during the testing stage.

5.6 Comparison Between Applied Soft Computing-Based Models

In the study, for forecasting the values of compressive strength of concrete utilized with waste foundry sand the advanced capabilities of the soft computing models are emphasized through a comparison shown in Table 4. The comparison indicates that both RF-based model and RT-based model exhibits similar accuracy during the training and testing stages, however in testing stage RF-based model outperformed other models that have been implemented in terms of forecasting the compressive strength with R as

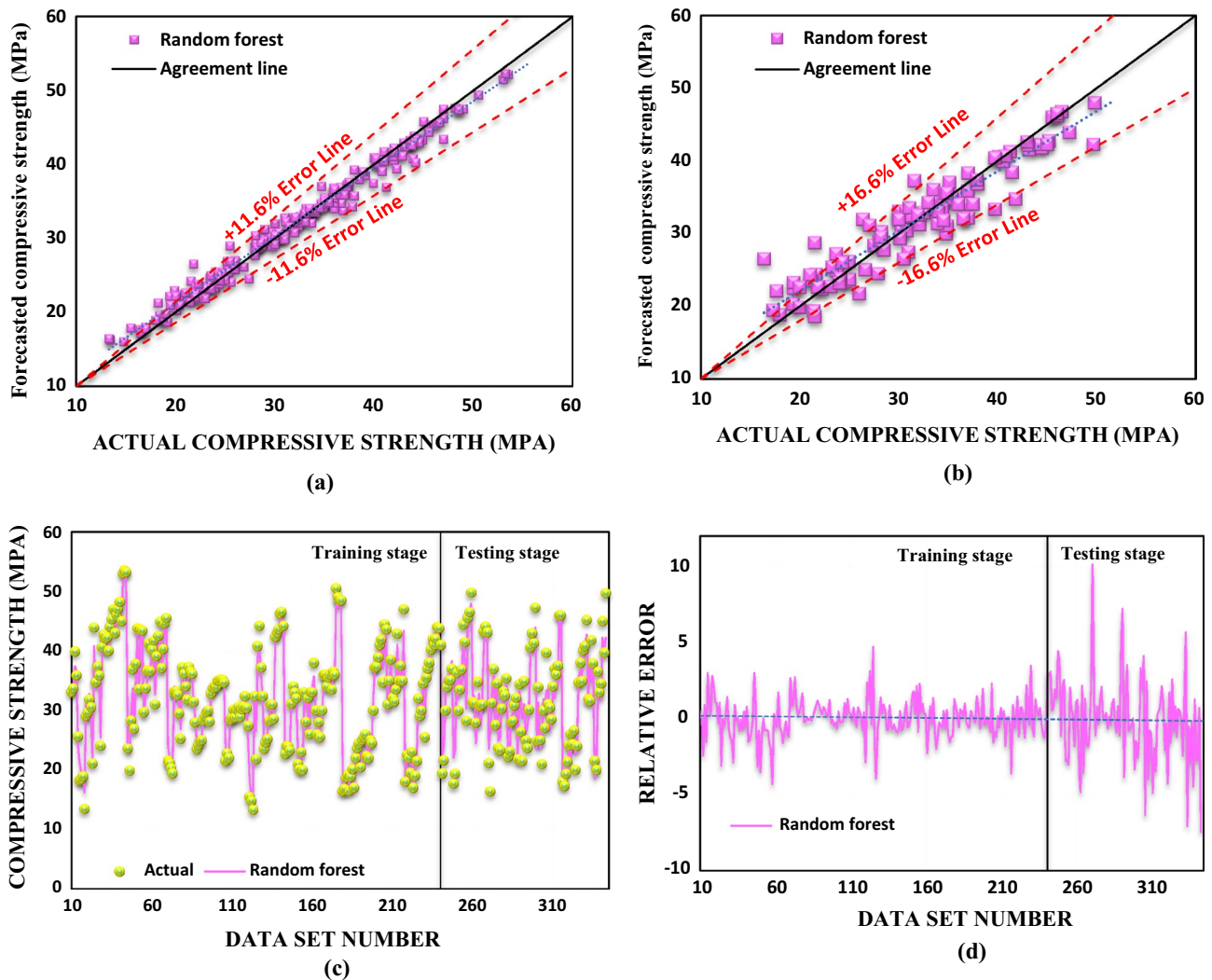


Fig. 13 Agreement plot of actual as well as forecasted values of MSP-based model: **a** direct correlation for the training stage ($n=240$) **b** direct correlation for the testing stage ($n=103$) **c** a head-to-head

agreement was executed concerning both the training and assessment stages ($n=343$) **d** training and testing stages' relative error ($n=343$)

0.99, 0.94, RMSE as 1.24, 2.79, MAE as 0.91, 2.09, RAE as 12.83%, 30.89%, RRSE as 14.24%, 33.89%, WI as 0.99, 0.96 and NSE 0.85, 0.88 for both the training as well as testing stages respectively.

According to the author's experiences, the performance of these models is not considerably impacted by increasing the amount of data set. Instead, it mostly results in a slight change to the values of the performance assessment statistics.

When analysing regression models including M5P, REP tree, ANN, RT, and RF, it has been noticed that the RF-based model gives superior outcomes during the testing

stage, characterized by a larger R -value and less errors in comparison with the other models. The performance assessment of all the models utilized during the testing and training stages is shown in Fig. 14a, b, recommending that the RF-based model's forecasted values are close to the actual data set and the error band for both training and testing lies between $\pm 33.3\%$.

Based on the descriptive statistics of actual and forecasted CS of concrete values during testing, is illustrated in Table 6, it is evident that the quartiles of the RF-based model most closely match the actual values. In addition, the actual values are in close agreement with the mean (third quartile) of the RF-based model. The RF-based model outperforms

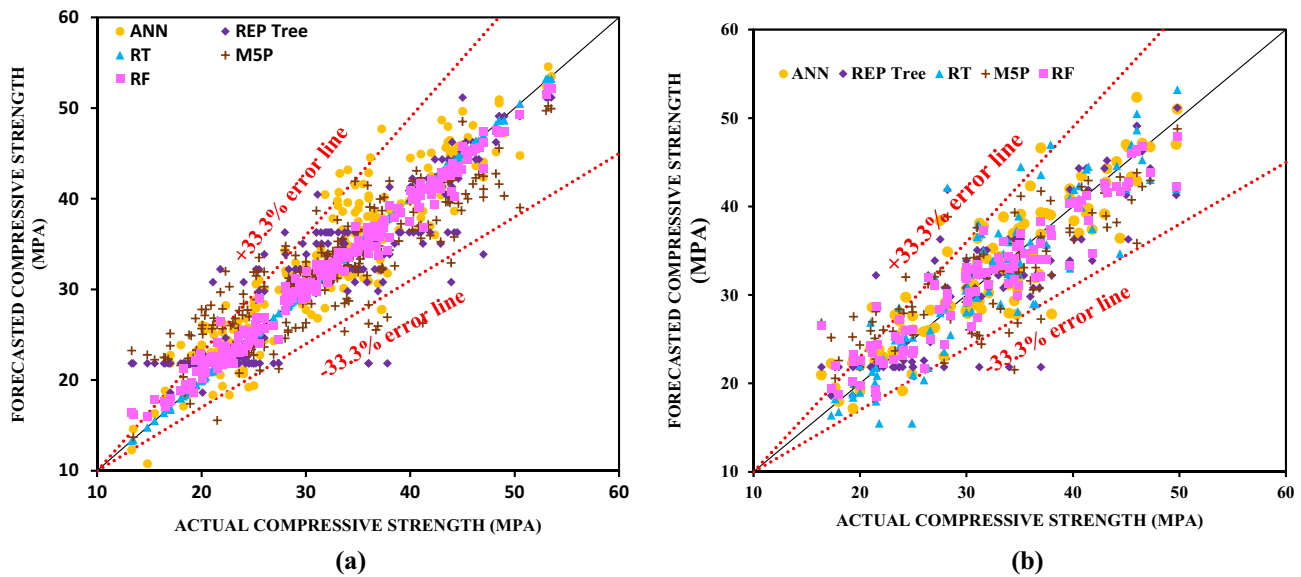


Fig. 14 Comparison between soft computing models: **a** training and **b** testing stage

Table 6 Descriptive statistics of observed and forecasted compressive strength values during testing

Statistic	Observed	M5P	REP tree	ANN	RT	RF
Testing stage						
Minimum	16.38	19.86	18.63	17.15	15.5	18.57
Maximum	49.8	48.79	51.17	52.36	53.23	47.97
1st Quartile	24.97	26.84	22.45	27.36	25.2	26.08
Mean	32	32.02	31.56	32.58	32.07	31.94
3rd Quartile	37	36.86	36.3	38.39	37.65	35.93

alternative established models in terms of their capability to predict the CS of concrete used with WFS during testing, as demonstrated by the following results. Additionally, an extensive statistical evaluation of all used models was done to see how close the forecasted and actual values were to each other. For training as well as testing stages, Box and Whisker graphs were plotted as presented in Fig. 15.

As shown in Fig. 16, Taylor diagrams were plotted graphically which represents the performance of all built models during the training and testing stages of the investigation. A thorough examination of the Taylor diagrams revealed that the sky-blue solid triangle representing the RT model closely aligns with the detected (black hollow) point in the training stage. In contrast, the pink solid point of the RF model is in closest proximity to the observed site during testing. This result suggests that the CS of the concrete utilized with waste foundry sand is precisely forecasted by both the RT-based and RF-based models.

In Fig. 13, the training (a) & testing (b) stages of the relationship between the observed CS dataset and

predictions provided by several soft computing methods are illustrated.

5.7 Sensitivity Analysis

In the final stage, sensitivity analysis technique was performed to evaluate the effect of independent parameters on the compressive strength (CS) as the dependent parameter. The findings of suggested sensitivity analysis (Fig. 17) indicates that ten independent parameters have fluctuating degree of effect on the CS. The curing period (CP) is the most influencing independent parameter in forecasting the CS and sand (S) has the least impact on the CS among ten independent parameters.

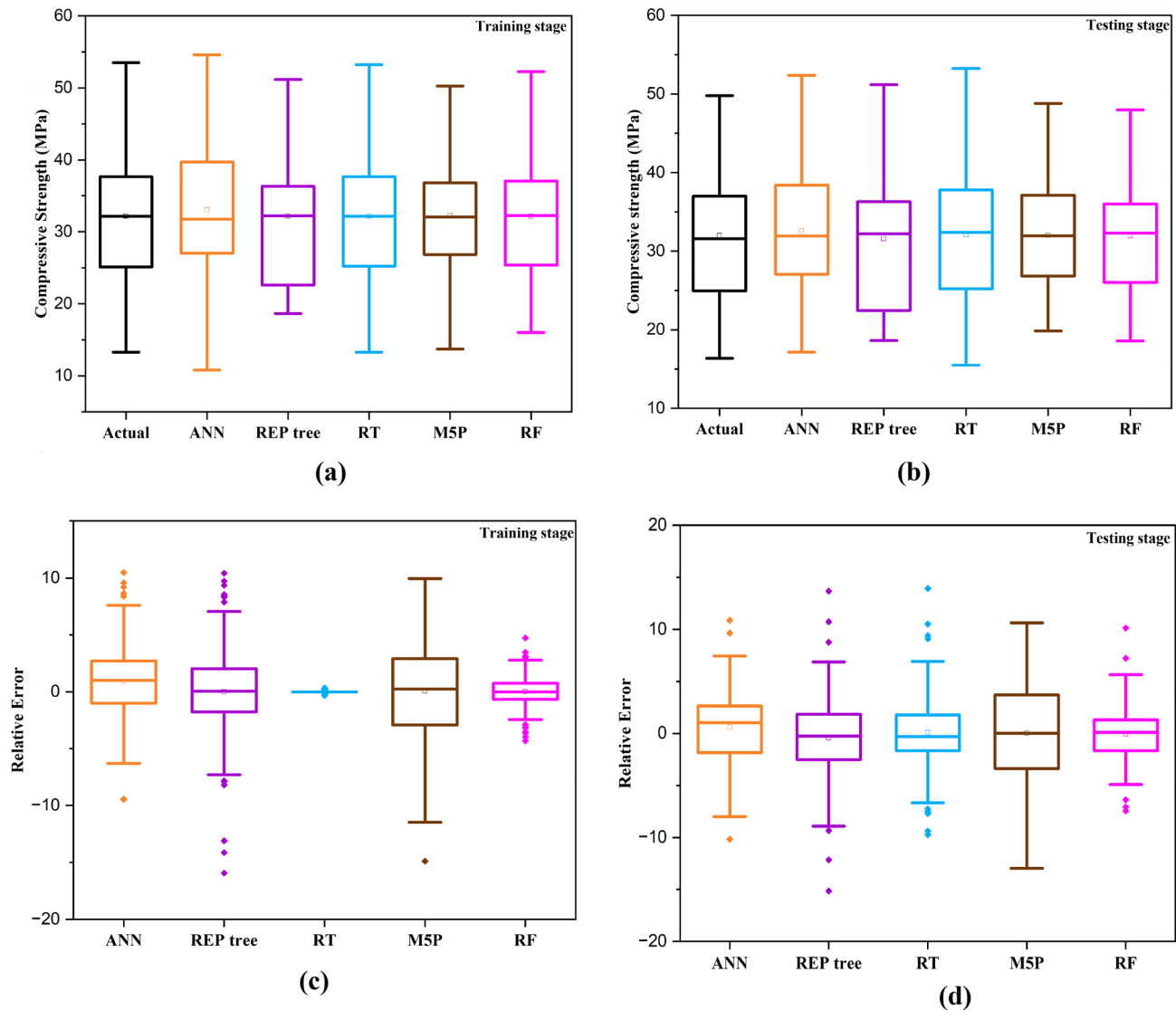


Fig. 15 Based on compressive strength (MPa) and the relative error, box and whisker plots were made for training stages (a, c) and the testing stages (b, d)

6 Conclusion

The predictions of CS values for concrete with waste foundry sand were obtained in this study using five soft computing methods: M5P, REP Tree ANN, RT, and RF model. Additionally, several performance evaluation metrics, such as R, RRSE, MAE, RMSE, WI, RAE, and NSE, have been used to compare the performance of these constructed models. Moreover, graphical techniques such as agreement plots, Taylor diagrams, and Box and Whisker plots have been employed to validate and verify the approximation performances of every soft computing method.

Statistical analysis indicated that the RF model beat other models, outperforming them by a margin of performance as indicated by the value of WI of 0.96, R of 0.94, MAE of 2.09, RMSE of 2.79, NSE of 0.88, RAE of 30.89%, and RRSE of 33.89% in the testing stage (Table 4).

However, the RT model exhibited competitive prediction abilities when associated with the ANN, M5P, and REP Tree models.

The results obtained from RF and RT models are promising hence further validation is required.

Ten main independent variables; CA, C, SL, FA, W, WFS, S, WFS/other, CP, and W/C were subjected to sensitivity analysis. The study highlighted the curing period

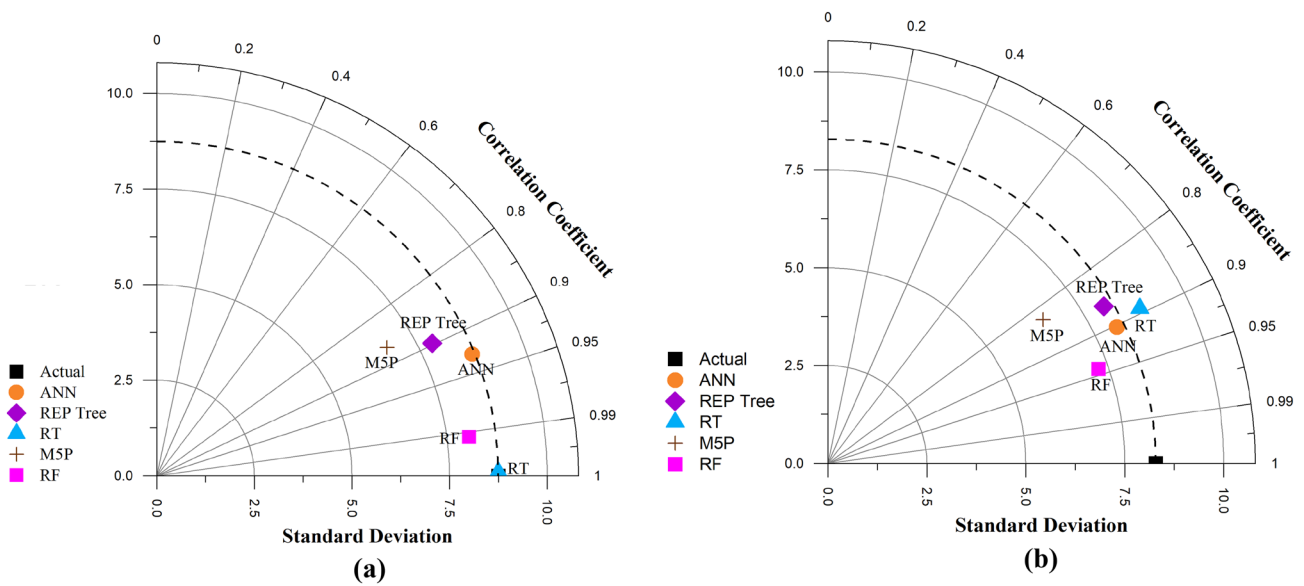


Fig. 16 Taylor diagrams for training stage **a** and testing stage **b**

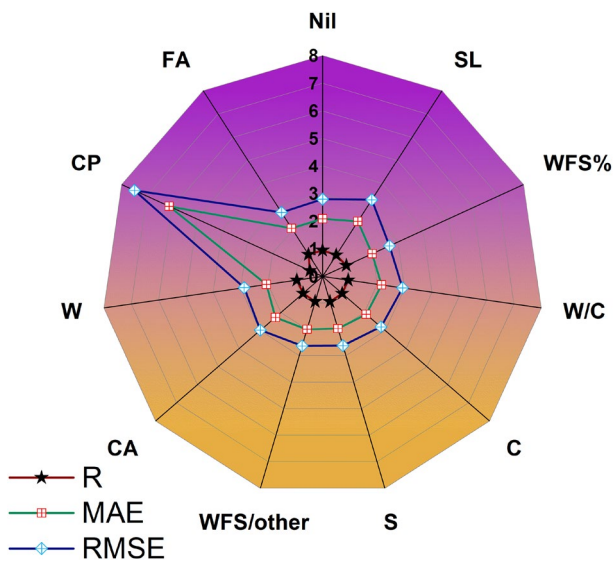


Fig. 17 Sensitivity analysis through radar plot

(CP) as the important factor in forecasting (CS) through the RF-based model. Consequently, it is imperative that the present investigation is validated by exploring CS behaviour and employing advanced soft computing tools. Hybrid approaches such as support vector regression with genetic algorithms, particle swarm optimization, adaptive neuro-fuzzy inference system, SVR with the firefly algorithm, SVR with PSO, and ANFIS with genetic algorithms illustrate the potential for additional research in the fusion of conventional methods and optimization techniques.

Data availability All data, models and code generated or used during the study paper in the submitted article.

Declarations

Conflict of Interest The authors declare that they have no competing interests.

References

- Dolage, D. A. R., Dias, M. G. S., & Ariyawansa, C. T. (2013). Offshore Sand as a Fine Aggregate for Concrete Production. *In Research Article British Journal of Applied Science & Technology*, 3(4), 13.
- Ray, S., Haque, M., Rahman, M. M., Sakib, M. N., & Al Rakib, K. (2021). Experimental investigation and SVM-based prediction of compressive and splitting tensile strength of ceramic waste aggregate concrete. *Journal of King Saud University - Engineering Sciences.*, 36(2), 112–221. <https://doi.org/10.1016/j.jksues.2021.08.010>
- Kumar, M., Biswas, R., Kumar, D. R., Samui, P., Kaloop, M. R., & Eldessouki, M. (2023). Soft computing-based prediction models for compressive strength of concrete. *Case Studies in Construction Materials*, 19, e02391. <https://doi.org/10.1016/j.cscm.2023.e02321>
- PK Mehta 2004 High-performance, high-volume fly ash concrete for sustainable development, in Proceedings of the International Workshop on Sustainable Development and Concrete Technology, Iowa State University
- Burhan, L., Ghafor, K., & Mohammed, A. (2019). Modeling the effect of silica fume on the compressive, tensile strengths, and durability of NSC and HSC in various strength ranges. *J. Build. Pathol. Rehabil.*, 4(1), 19.
- Siddique, R., & Noumowe, A. (2008). Utilization of spent foundry sand in controlled low strength materials and concrete. *Resour. Conserv. Recycle.*, 53, 27–35. <https://doi.org/10.1016/j.resconrec.2008.09.007>

7. Topçu, I. B., & Sarıdemir, M. (2008). Prediction of compressive strength of concrete containing fly ash using artificial neural networks and fuzzy logic. *Computational Materials Science*, 41(3), 305–311. <https://doi.org/10.1016/j.commatsci.2007.04.009>
8. Alyousef, R., Nassar, R. U. D., Khan, M., Arif, K., Fawad, M., Hassan, A. M., & Ghamry, N. A. (2023). Forecasting the strength characteristics of concrete incorporating waste foundry sand using advance machine algorithms including deep learning. *Case Studies in Construction Materials*, 19, e02459. <https://doi.org/10.1016/J.CSCM.2023.E02459>
9. Behnood, A., & Golafshani, E. M. (2020). Machine learning study of the mechanical properties of concretes containing waste foundry sand. *Construction and Building Materials*, 243, 118152. <https://doi.org/10.1016/J.CONBUILDMAT.2020.118152>
10. Deng, Z. (2022). Utilisation of steel fibres to reinforce waste glass concrete: Alkali–silica reaction, engineering properties, and 3D mesoscale modelling. *Case Studies in Construction Materials*, 17, e01686.
11. Deng, Z. (2022). Utilization of lithium nitrate to mitigate alkali–silica reaction of architectural glass mortar: Characteristics and mechanisms. *Construction and Building Materials*, 315, 125433.
12. Deng, Z., Yang, Z., & Pan, X. (2023). Synergetic effects of recycled crumb rubber and glass cullet on the engineering properties of geopolymer mortar. *Cement and Concrete Composites*, 137, 104907.
13. Deng, Z., Zhang, S., & Deng, Z. (2023). PVA fiber-reinforced geopolymer mortar made with hybrid recycled aggregates: Toward thermal insulation, lightweight and improved durability. *Journal of Cleaner Production*, 426, 139200.
14. Deng, Z., Yang, Z., Bian, J., Pan, X., Wu, G., Guo, F., & Chen, S. (2022). Engineering properties of PVA fibre-reinforced geopolymer mortar containing waste oyster shells. *Materials*, 15(19), 7013.
15. Aggarwal, Y., & Siddique, R. (2014). Microstructure and properties of concrete using bottom ash and waste foundry sand as partial replacement of fine aggregates. *Construction and Building Materials*, 54, 210–223. <https://doi.org/10.1016/j.conbuildmat.2013.12.051>
16. Prabhu, G. G., Hyun, J. H., & Kim, Y. Y. (2014). Effects of foundry sand as a fine aggregate in concrete production. *Construction and Building Materials*, 70, 514–521. <https://doi.org/10.1016/j.conbuildmat.2014.07.070>
17. Anagnostopoulos, A., Elena Navarro, M., Sharma, S., Ahmad, A., Maksum, Y., & Ding, Y. (2024). From waste to value: Utilising waste foundry sand in thermal energy storage as a matrix material in composites. *Solar Energy*, 268, 112294. <https://doi.org/10.1016/j.solener.2023.112294>
18. Bhardwaj, B., & Kumar, P. (2017). Waste foundry sand in concrete: A review. In *Construction and Building Materials*, 156, 661–674. <https://doi.org/10.1016/j.conbuildmat.2017.09.010>
19. Kaur, G., Siddique, R., & Rajor, A. (2012). Properties of concrete containing fungal-treated waste foundry sand. *Construction and Building Materials*, 29, 82–87. <https://doi.org/10.1016/j.conbuildmat.2011.08.091>
20. Singh, G., & Siddique, R. (2012). Abrasion resistance and strength properties of concrete containing waste foundry sand (WFS). *Construction and Building Materials*, 28(1), 421–426. <https://doi.org/10.1016/j.conbuildmat.2011.08.087>
21. Singh, G., & Siddique, R. (2012). Effect of waste foundry sand (WFS) as partial replacement of sand on the strength, ultrasonic pulse velocity and permeability of concrete. *Construction and Building Materials*, 26(1), 416–422. <https://doi.org/10.1016/j.conbuildmat.2011.06.041>
22. Naik, T. R., Patel, V. M., Parikh, D. M., & Tharaniy, M. P. (1994). Utilization of used foundry sand in concrete. *Journal of Materials in Civil Engineering*, 6(2), 254–263.
23. Basar, H. M., & Aksoy, D. N. (2012). The effect of waste foundry sand (WFS) as partial replacement of sand on the mechanical, leaching and micro-structural characteristics of ready-mixed concrete. *Construction and Building Materials*, 35, 508–515. <https://doi.org/10.1016/j.conbuildmat.2012.04.078>
24. Khatib, J. M., Herki, B. A., & Kenai, S. (2013). Capillarity of concrete incorporating waste foundry sand. *Construction and Building Materials*, 47, 867–871. <https://doi.org/10.1016/j.conbuildmat.2013.05.013>
25. Khatib, J. M., Profile, S., Bougara, A., Negim, E.-S., & Kenai, S. (2012). Utilisation of Foundry Sand in Concrete Production. *The University of Wisconsin Milwaukee Centre for By products Utilization*, 14(1), 244–256.
26. Siddique, R., Singh, G., Belarbi, R., Ait-Mokhtar, K., & Kunal. (2015). Comparative investigation on the influence of spent foundry sand as partial replacement of fine aggregates on the properties of two grades of concrete. *Construction and Building Materials*, 83, 216–222. <https://doi.org/10.1016/j.conbuildmat.2015.03.011>
27. Mohammed, A., Rafiq, S., Sihag, P., Kurda, R., & Mahmood, W. (2021). Soft computing techniques: Systematic multiscale models to predict the compressive strength of HVFA concrete based on mix proportions and curing times. In *Journal of Building Engineering*, 33, 101851. <https://doi.org/10.1016/j.job.2020.101851>
28. Khasawneh, M. A., Al-Akhrass, H. I., Rabab'ah, S. R., & Al-sugaiyer, A. O. (2024). Prediction of California bearing ratio using soil index properties by regression and machine-learning techniques. *International Journal of Pavement Research and Technology*, 17(2), 306–324.
29. Suthar, M., & Aggarwal, P. (2018). Predicting CBR value of stabilized pond ash with lime and lime sludge using ANN and MR models. *Int J Geosynth Ground Eng*, 4(1), 1–7. <https://doi.org/10.1007/s40891-017-0125-3>
30. Suthar, M., & Aggarwal, P. (2019). Modeling CBR value using RF and MSP techniques. *Mendel*, 25(1), 7378.
31. Khattai, J., & Grover, K. S. (2023). CBR prediction of pavement materials in unsoaked condition using LSSVM, LSTM-RNN, and ANN approaches. *International Journal of Pavement Research and Technology*, 17, 750–787.
32. Sihag, P., Suthar, M., & Mohanty, S. (2019). Estimation of UCS-FT of dispersive soil stabilized with fly ash, cement clinker and GGBS by artificial intelligence. *Iranian J Sci Technol Trans Civil Eng*, 45(2), 901–912. <https://doi.org/10.1007/s40996-019-00329-0>
33. Suthar, M. (2020). Modeling of UCS value of stabilized pond ashes using adaptive neuro-fuzzy inference system and artificial neural network. *Soft Computing*, 24, 14561–14575. <https://doi.org/10.1007/s00500-020-04806-x>
34. Suthar, M. (2020). Applying several machine learning approaches for prediction of unconfined compressive strength of stabilized pond ashes. *Neural Computing and Applications*, 32(13), 9019–9028. <https://doi.org/10.1007/s00521-019-04411-6>
35. Cary, C. E., & Zapata, C. E. (2011). Resilient modulus for unsaturated unbound materials. *Road Mater Pavement Des*, 12(3), 615–638. <https://doi.org/10.1080/14680629.2011.9695263>
36. Jalota, S., & Suthar, M. (2023). Prediction of Marshall stability of asphalt concrete reinforced with polypropylene fibre using different soft computing techniques. *Soft Computing*, 14, 2612. <https://doi.org/10.1007/s00500-023-08339-x>
37. Upadhya, A., Thakur, M. S., Sharma, N., & Sihag, P. (2022). Assessment of soft computing-based techniques for the prediction of marshall stability of asphalt concrete reinforced with glass fiber. *International Journal of Pavement Research and Technology*, 15(6), 1366–1385.

38. Ali, I., & Suthar, M. (2023). Comparison between Random forest and M5P to predict the compressive strength of concrete modified with solid wastes. *IOP conference series: earth and environmental science*, 110(1), 012085. <https://doi.org/10.1088/1755-1315/1110/1/012085>
39. Calis, G., Yildizel, S. A., & Keskin, U. S. (2023). Predicting Compressive Strength of Color Pigment Incorporated Roller Compacted Concrete via Machine Learning Algorithms: A Comparative Study. *International Journal of Pavement Research and Technology*, 23, 1–17.
40. Choudhary, J., Kumar, B., & Singh, S. (2021). Assessment of engineering and environmental suitability of waste bituminous concrete containing waste biomass ash. *International Journal of Pavement Research and Technology*, 14, 751–763.
41. MA Khan M Suthar 2023 Assessing the results of compressive strength of ultra high-performance concrete using soft computing. In: *IOP Conference series: earth and environmental science*, <https://doi.org/10.1088/1755-1315/1110/1/012089>
42. Rani, K., Suthar, M., Sihag, P., & Boora, A. (2021). Experimental investigation and prediction of strength development of GGBFS-, LFSand SCBA-based green concrete using soft computing techniques. *Arabian Journal of Geosciences*, 14(23), 1–15.
43. Pakdel, P., JamshidiChenari, R., & Veiskarami, M. (2019). An estimate of the bearing capacity of shallow foundations on anisotropic soil by limit equilibrium and soft computing technique. *Geomech Geoeng*, 1, 1342.
44. Lawal, A. I., & Kwon, S. (2021). Application of artificial intelligence to rock mechanics: An overview. *J Rock Mech Geotech Eng*, 13(1), 248–266. <https://doi.org/10.1016/j.jrmge.2020.05.010>
45. Shahrour, I., & Zhang, Z. (2021). Use of soft computing techniques for tunnelling optimization of tunnel boring machines. *Underground Space*, 6(3), 233–239. <https://doi.org/10.1016/j.undsp.2019.12.001>
46. Sharma, S. K., Sharma, V., Mohamed, H. I., Khan, H., & Ahmed, S. S. (2021). Supervise the physicochemical quality of ground water using soft computing technique. *Environmental Technology*, 45(11), 2099–2107. <https://doi.org/10.1080/09593330.2021.1983023>
47. Raja, M. N. A., Shukla, S. K., & Khan, M. U. A. (2021). An intelligent approach for predicting the strength of geosynthetic-reinforced subgrade soil. *International Journal of Pavement Engineering*, 23(10), 3505–3521. <https://doi.org/10.1080/10298436.2021.1904237>
48. Ahmed, H. U., Abdalla, A. A., Mohammed, A. S., & Mohammed, A. A. (2022). Mathematical modelling techniques to predict the compressive strength of high-strength concrete incorporated metakaolin with multiple mix proportions. *Cleaner Materials*, 5, 100132. <https://doi.org/10.1016/j.clema.2022.100132>
49. Farooq, F., Czarnecki, S., Niewiadomski, P., Aslam, F., Alabduljabbar, H., Ostrowski, K. A., Śliwa-Wieczorek, K., Nowobilski, T., & Malazdrewicz, S. (2021). A comparative study for the prediction of the compressive strength of self-compacting concrete modified with fly ash. *Materials*, 14(17), 3390. <https://doi.org/10.3390/ma14174934>
50. Shah, H. A., Yuan, Q., Akmal, U., Shah, S. A., Salmi, A., Awad, Y. A., Shah, L. A., Iftikhar, Y., Javed, M. H., & Khan, M. I. (2022). Application of machine learning techniques for predicting compressive, splitting tensile, and flexural strengths of concrete with metakaolin. *Materials*, 15(15), 5435. <https://doi.org/10.3390/ma15155435>
51. Mehta, V. (2023). Machine learning approach for predicting concrete compressive, splitting tensile, and flexural strength with waste foundry sand. *Journal of Building Engineering*, 70, 106363. <https://doi.org/10.1016/j.jobbe.2023.106363>
52. Manoharan, T., Lakshmanan, D., Mylsamy, K., Sivakumar, P., & Sircar, A. (2018). Engineering properties of concrete with partial utilization of used foundry sand. *Waste Management*, 71, 454–460. <https://doi.org/10.1016/j.wasman.2017.10.022>
53. Reshma, T. V., Manjunatha, M., Sankalparsi, S., & Tanu, H. M. (2021). Effect of waste foundry sand and fly ash on mechanical and fresh properties of concrete. *Materials Today: Proceedings*, 47, 3625–3632. <https://doi.org/10.1016/j.matpr.2020.12.821>
54. Zai, R. A. A., & Salhotra, S. (2020). Effect of waste foundry sand and glass fiber on mechanical properties and fire resistance of high-strength concrete. *Materials Today: Proceedings*, 33, 1733–1740. <https://doi.org/10.1016/j.matpr.2020.08.321>
55. Siddique, R., Singh, G., & Singh, M. (2018). Recycle option for metallurgical by-product (Spent Foundry Sand) in green concrete for sustainable construction. *Journal of Cleaner Production*, 172, 1111–1120. <https://doi.org/10.1016/j.jclepro.2017.10.255>
56. Parashar, A., Aggarwal, P., Saini, B., Aggarwal, Y., & Bishnoi, S. (2020). Study on performance enhancement of self-compacting concrete incorporating waste foundry sand. *Construction and Building Materials*, 251. <https://doi.org/10.1016/j.conbuildmat.2020.118875>
57. Mavroulidou, M., & Lawrence, D. (2018). Can waste foundry sand fully replace structural concrete sand? *Journal of Material Cycles and Waste Management*, 21(3), 594–605. <https://doi.org/10.1007/s10163-018-00821-1>
58. Prasad, V. D., Prakash, E. L., Abishek, M., Ushanth Dev, K., & Sanjay Kiran, C. K. (2018). Study on concrete containing Waste Foundry Sand, Fly Ash and Polypropylene fibre using Taguchi Method. *Materials Today: Proceedings*, 5(11), 23964–23973. <https://doi.org/10.1016/j.matpr.2018.10.189>
59. Bilal, H., Yaqub, M., Ur Rehman, S. K., Abid, M., Alyousef, R., Alabduljabbar, H., & Aslam, F. (2019). Performance of foundry sand concrete under ambient and elevated temperatures. *Materials*, 12(16). <https://doi.org/10.3390/ma12162645>
60. Mushtaq, S. M., Siddique, R., Goyal, S., & Kaur, K. (2021). Experimental studies and drying shrinkage prediction model for concrete containing waste foundry sand. *Cleaner Engineering and Technology*, 2. <https://doi.org/10.1016/j.clet.2021.100071>
61. Sowmya M., & Kumar J. D. C. (2015). Mixing of waste foundry sand in concrete. *International Journal of Engineering Research and Science & Technology*, 4(4).
62. Thiruvengadam, M., Pandian, S., Santra, M., & Subramanian, D. (2020). Use of waste foundry sand as a partial replacement to produce green concrete: Mechanical properties, durability attributes and its economical assessment. *Environmental Technology and Innovation*, 19. <https://doi.org/10.1016/j.eti.2020.101022>
63. Kavitha, O. R., Shyamala, G., & Akshana, V. (2021). Study of sustainable concrete property containing waste foundry sand. *Materials Today: Proceedings*, 39, 855–860. <https://doi.org/10.1016/J.MATPR.2020.10.359>
64. Siddique, R., Schutter, G. de, & Noumowe, A. (2009). Effect of used-foundry sand on the mechanical properties of concrete. *Construction and Building Materials*, 23(2), 976–980. <https://doi.org/10.1016/j.conbuildmat.2008.05.005>
65. Mehmannaavaz, T., Khalilikhorrani, V., Sajjadi, S. M., & Samadi, M. (2014). Presenting an appropriate neural network for optimal mix design of roller compacted concrete dams. *Research Journal of Applied Sciences, Engineering and Technology*, 7(9), 1872–1877.
66. Khan, M. I. (2012). Predicting properties of High Performance Concrete containing composite cementitious materials using Artificial Neural Networks. *Automation in Construction*, 22, 516–524. <https://doi.org/10.1016/j.autcon.2011.11.011>
67. Sobhani, J., Najimi, M., Pourkhorshidi, A. R., & Parhizkar, T. (2010). Prediction of the compressive strength of no-slump concrete: A comparative study of regression, neural network and ANFIS models. *Construction and Building Materials*, 24(5), 709–718. <https://doi.org/10.1016/j.conbuildmat.2009.10.037>

68. Breiman, L. (1996). Bagging predictors. *Machine Learning*, 24(2), 123–140. <https://doi.org/10.1007/BF00058655>
69. Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
70. Wright, M. N., & Ziegler, A. (2017). Ranger: A fast implementation of random forests for high dimensional data in C++ and R. *Journal of Statistical Software*, 77(1), 1508.
71. Ramesh, D., Pasha, S., & Roopa, G. (2017). A comparative analysis of classification algorithms on weather dataset using data mining tool. *Oriental Journal of Computer Science and Technology*, 10(4), 788–792.
72. Kalmegh S 2015. Analysis of WEKA Data Mining Algorithm REPTree, Simple Cart and RandomTree for Classification of Indian News. In *IJISSET-International Journal of Innovative Science, Engineering & Technology* 2:2. www.ijiset.com
73. Ajmera, T. K., & Goyal, M. K. (2012). Development of stage-discharge rating curve using model tree and neural networks: An application to Peachtree Creek in Atlanta. *Expert Systems with Applications*, 39(5), 5702–5710. <https://doi.org/10.1016/j.eswa.2011.11.101>
74. Onyari, E., & Ilunga, F. (2013). Application of MLP neural network and M5P model tree in predicting streamflow: A case study of Luvuvhu catchment, South Africa. *Int J Innov Manag Technol*, 4, 11.
75. Deshpande, N., Londhe, S., & Kulkarni, S. (2014). Modeling compressive strength of recycled aggregate concrete by Artificial Neural Network, Model Tree and Non-linear Regression. *International Journal of Sustainable Built Environment*, 3(2), 187–198. <https://doi.org/10.1016/j.ijsbe.2014.12.002>
76. Nikoo, M. R., Karimi, A., Kerachian, R., Poorsepahy-Samian, H., & Daneshmand, F. (2013). Rules for optimal operation of reservoir-river-groundwater systems considering water quality targets: application of M5P model. *Water Resources Management*, 27(8), 2771–2784. <https://doi.org/10.1007/s11269-013-0314-3>
77. Bassar Quinlan, J. R. (1992). *Learning with continuous classes*. World Scientific.
78. Aldous, D. (1993). The continuum random tree III. *The Annals of Probability*, 21(1), 248–289.
79. Swe, T. T. (2019). Analysis of Tree Based Supervised Learning Algorithms on Medical Data. *International Journal of Scientific and Research Publications (IJSRP)*, 9(4), 8817.
80. Singh, B., Sihag, P., & Singh, K. (2018). Comparison of infiltration models in NIT Kurukshetra campus. *Applied Water Science*, 8(2), 63. <https://doi.org/10.1007/s13201-018-0708-8>
81. Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer Science & Business Media. <https://doi.org/10.1007/s12517-021-08869-4>

author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.



Suhaib Rasool Wani is a M.Tech Scholar in Transportation Engineering from Department of Civil Engineering, at Chandigarh University, Punjab, India. He has received his B.Tech. from Chandigarh University, Punjab, India. His research interest includes application of soft computing techniques like artificial neural networks, linear regression, tree- and forest-based algorithms, etc. in transportation projects.



Manju Suthar is an assistant professor of Transportation Engineering, Department of Civil Engineering at Chandigarh University, Punjab, India. She received her B.Tech. degree in Civil Engineering from Lovely Professional University, Punjab, India, in 2011. She also obtained her M.Tech. and Ph.D. degrees in Transportation Engineering from National Institute of Technology, Kurukshetra, India, in 2013 and 2018, respectively. Her research interests includes material Testing and application of mathematical modeling, linear/nonlinear regression analysis-based modeling, artificial neural networks, adaptive neuro-fuzzy inference systems, support vector machine, Gaussian process, tree- and forest-based algorithms, etc. in Civil Engineering projects. She has published various national and international papers related to highway and construction engineering.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the